

Код программы

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class Department():
    def __init__(self, id, name, tuitionFee, facultyId):
        self.id = id
        self.name = name
        self.tuitionFee = tuitionFee
        self.facultyId = facultyId

class Faculty():
    def __init__(self, id, name):
        self.id = id
        self.name = name

class FacultyDepartment():
    def __init__(self, facultyId, departmentId):
        self.facultyId = facultyId
        self.departmentId = departmentId

def JoinFacultyToDepartmentIfOneToManyRelationship(departments, faculties):
    one_to_many = [
        (department.name, department.tuitionFee, faculty.name)
        for faculty in faculties
        for department in departments
        if department.facultyId == faculty.id
    ]

    return one_to_many

def JoinFacultyToDepartmentIfManyToManyRelationship(departments, faculties,
facultyDepartments):
    many_to_many_temp = [
        (faculty.name, facultyDepartment.facultyId, facultyDepartment.departmentId)
        for faculty in faculties
        for facultyDepartment in facultyDepartments
        if facultyDepartment.facultyId == faculty.id
    ]

    many_to_many = [
        (department.name, department.tuitionFee, facultyName)
        for facultyName, _, departmentId in many_to_many_temp
        for department in departments
        if department.id == departmentId
    ]

    return many_to_many

def PrintTaskNumber(number):
    print(f'\nЗадание {number}')

def PerformFirstRequest(one_to_many):
    PrintTaskNumber('B1')
    resB1 = list(filter(lambda row: row[0].startswith('И'), one_to_many))

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print(*resB1, sep='\n')

def PerformSecondTask(faculties, one_to_many):
    PrintTaskNumber('B2')

    resA2Unsorted = []

    for faculty in faculties:
        facultyDepartments = list(filter(lambda row: row[2] == faculty.name,
one_to_many))
        if facultyDepartments:
            facultyTuitionFees = [row[1] for row in facultyDepartments]
            resA2Unsorted.append((faculty.name, min(facultyTuitionFees)))

    resB2 = sorted(resA2Unsorted, key=lambda row : row[1])
    print(*resB2, sep='\n')

def PerformThirdTask(many_to_many):
    PrintTaskNumber('B3')
    resA3 = sorted(many_to_many, key=lambda row : row[0])
    print(*resA3, sep='\n')

def main():
    faculties = [
        Faculty(1, 'Информатика и системы управления'),
        Faculty(2, 'Специальное машиностроение'),
        Faculty(3, 'Инженерный бизнес и менеджмент'),

        Faculty(11, '(Другие) Информатика и системы управления'),
        Faculty(12, '(Другое) Специальное машиностроение'),
        Faculty(13, '(Другой) Инженерный бизнес и менеджмент'),
    ]

    departments = [
        Department(1, 'ИУ5', 449000, 1),
        Department(2, 'СМ12', 539000, 2),
        Department(3, 'ИУ7', 549000, 1),
        Department(4, 'ИБМЗ', 439000, 3),
        Department(5, 'СМ7', 459000, 2)
    ]

    facultyDepartments = [
        FacultyDepartment(1, 1),
        FacultyDepartment(2, 2),
        FacultyDepartment(1, 3),
        FacultyDepartment(3, 4),
        FacultyDepartment(2, 5),

        FacultyDepartment(11, 1),
        FacultyDepartment(12, 2),
        FacultyDepartment(11, 3),
        FacultyDepartment(13, 4),
        FacultyDepartment(12, 5)
    ]

```

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]

one_to_many = JoinFacultyToDepartmentIfOneToManyRelationship(departments, faculties)
many_to_many = JoinFacultyToDepartmentIfManyToManyRelationship(departments,
faculties, facultyDepartments)

PerformFirstRequest(one_to_many)
PerformSecondTask(faculties, one_to_many)
PerformThirdTask(many_to_many)

if __name__ == "__main__":
    main()

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Результат работы программы

● proshchaevalexey@MacBook-Air-Aleksej-2 rk_1 % python3 main.py

Задание B1

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('ИУ5', 449000, 'Информатика и системы управления')
('ИУ7', 549000, 'Информатика и системы управления')
('ИБМЗ', 439000, 'Инженерный бизнес и менеджмент')

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Задание B2

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('Инженерный бизнес и менеджмент', 439000)
('Информатика и системы управления', 449000)
('Специальное машиностроение', 459000)

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Задание B3

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('ИБМЗ', 439000, 'Инженерный бизнес и менеджмент')
('ИБМЗ', 439000, '(Другой) Инженерный бизнес и менеджмент')
('ИУ5', 449000, 'Информатика и системы управления')
('ИУ5', 449000, '(Другие) Информатика и системы управления')
('ИУ7', 549000, 'Информатика и системы управления')
('ИУ7', 549000, '(Другие) Информатика и системы управления')
('СМ12', 539000, 'Специальное машиностроение')
('СМ12', 539000, '(Другое) Специальное машиностроение')
('СМ7', 459000, 'Специальное машиностроение')
('СМ7', 459000, '(Другое) Специальное машиностроение')

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○ proshchaevalexey@MacBook-Air-Aleksej-2 rk_1 %